PROJECT REPORT

Global Fuel Prices and Taxation Trends

A Comprehensive Global Performance Intelligence Report

	Information
Project Title	Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report
Student / Team	Individual Project
Team ID	N/A – Individual Project
Date	1 November 2026
Platform	Tableau Public
Data Source	global_fuel_prices_Dataset.csv

FROM RAW RECORDS TO GLOBAL INSIGHT

This report documents the project journey: understanding the problem, shaping the analysis, designing the Tableau solution, planning the work, and reviewing the implemented dashboard and Story.

Published project: **Open the Tableau Public Story**

Project focus

Explore how petrol, diesel and LPG prices, Brent crude trends and fuel taxation differ across countries, regions and years. The report is designed to make comparisons easier through maps, KPI cards, trend charts and a guided Tableau Story.

1. INTRODUCTION

1.1 Project Overview

Fuel prices influence household spending, transport costs, trade and economic planning. However, fuel-price and tax information can be difficult to interpret when it is spread across many countries, regions, years and fuel categories. This project uses a global fuel-price CSV dataset and Tableau Public to bring key comparisons into one visual report.

The dashboard examines average petrol, diesel and LPG prices, the yearly movement of these fuel prices, Brent crude trends and differences in fuel-tax percentages. It combines geographic views with time-series and ranking charts so that users can move from a broad global picture to focused country comparisons.

1.2 Purpose

- Summarize average petrol, diesel and LPG prices and average tax percentage through KPI cards.
- Compare petrol and diesel prices geographically using country maps.
- Explore yearly trends for petrol, diesel, LPG and Brent crude.
- Compare average tax percentage across regions and identify the five countries with the highest average tax percentage.
- Compare petrol, diesel and LPG prices across Canada, Mexico and the United States.

Intended users include students, researchers, energy analysts and decision-makers who need a clear starting point for exploring country-level fuel-price and taxation patterns.

2. IDEATION PHASE

2.1 Problem Statement

Global fuel-price and taxation records contain many observations across locations, time periods and fuel types. Comparing these records manually can be time-consuming and can make important patterns difficult to spot. Users need a consolidated visual view that allows them to compare countries and regions, review yearly movements and identify differences in taxation.

How might we make global fuel-price and tax information easier to compare and understand through a single, accessible visual analytics experience?

2.2 Empathy Map Canvas

Empathy area	Project interpretation
Says	“I need to compare fuel prices and taxes without searching through many separate records.”
Thinks	Which countries and regions have higher average prices? How have prices changed over time?
Does	Filters data by country or year, checks maps and charts, and compares fuel types and tax percentages.
Feels	May feel overwhelmed by raw records; clearer visual comparisons can make the analysis easier to follow.
Pains	Scattered data, large tables, inconsistent labels or units, and time-consuming manual comparisons.
Gains	A quick overview, visible trends, geographic comparison and a guided explanation of findings.

This empathy map is a project-design assumption describing likely users. It is not presented as evidence from formal customer interviews.

2.3 Brainstorming

- Use KPI cards to give an immediate summary of the main measures.
- Use country maps to make geographic price differences visible.
- Use yearly line charts to show changes in fuel and Brent crude prices.
- Use a regional pie chart and Top 5 bar chart to communicate tax comparisons.
- Use a focused three-country view to compare Canada, Mexico and the United States.
- Arrange the visualizations as a Tableau Story so viewers can follow a logical sequence.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User activity	Need / project response
Discover	Opens the published Tableau Story.	Clear title, context and accessible public link.
Explore	Reviews KPI cards and country maps.	Quick summary and geographic comparison.
Compare	Checks yearly trends, regions and selected countries.	Consistent AVG measures and available filters.
Interpret	Looks for notable differences and patterns.	Readable labels, legends and chart titles.
Share	Uses the Story for discussion or presentation.	A guided sequence and shareable Tableau Public link.

3.2 Solution Requirement

- Connect the global fuel prices CSV dataset to Tableau Public.
- Use Country, Region and Year as dimensions and fuel-price/tax measures as numeric fields.
- Create average measures for petrol, diesel, LPG, Brent crude and tax percentage.
- Build KPI cards, country maps, yearly trends, tax comparison charts and a selected-country comparison.
- Combine the relevant worksheets into a dashboard and a seven-point Tableau Story.
- Publish the Story on Tableau Public for browser-based viewing.

3.3 Data Flow Diagram

Data moves from the CSV source into Tableau Public, where fields are checked and measures are aggregated. The resulting worksheets are assembled into a dashboard and Story for users to explore.

3.4 Technology Stack

Layer	Technology / resource	Role
Data source	CSV file: global_fuel_prices_Dataset.csv	Stores the global fuel-price and taxation records used for analysis.
Analytics and visualization	Tableau Public	Connects to the CSV, aggregates measures and builds worksheets, dashboards and Story.
Access	Web browser / Tableau Public	Allows users to view the published project.
Other technologies	No SQL database, Snowflake warehouse or Streamlit app used in the implemented version.	Not part of the current project implementation.

4. PROJECT DESIGN

4.1 Problem–Solution Fit

The problem is the difficulty of understanding fuel-price and tax differences from a large, multi-country dataset. The proposed fit is a single Tableau Public report that combines summary indicators with geographic, time-based and tax-focused views. It supports exploration and comparison; it does not replace official market or policy data sources.

User need	Tableau response
Quick overview	Four KPI cards for average diesel, LPG, petrol and tax percentage.
Geographic comparison	Petrol and diesel price maps by country.
Time comparison	Yearly trends for petrol, diesel, LPG and Brent crude.
Tax comparison	Average tax percentage by region and Top 5 countries by tax percentage.
Focused country comparison	Petrol, diesel and LPG comparison for Canada, Mexico and the United States.

4.2 Proposed Solution

The proposed solution is an interactive Tableau Public dashboard supported by a seven-point Story. It uses average aggregations to make measures comparable and organizes the views so that users can first see the global pattern, then examine trends and taxation, and finally review a focused country comparison.

4.3 Solution Architecture

- Input: CSV dataset containing country, region, year, fuel-price and taxation fields.
- Preparation: inspect field names and data types; use geographic and time dimensions appropriately.
- Analysis: calculate average fuel prices and average tax percentage using Tableau aggregations.
- Presentation: build KPI cards, maps, trend charts, tax charts and country comparison.
- Delivery: publish the dashboard and Story on Tableau Public.

The current implementation is CSV-to-Tableau. Snowflake, SQL pipelines, automated ETL and Streamlit deployment are not claimed as implemented components.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was organized into four planned workstreams/sprints. The sequence moves from understanding the data to creating visualizations, assembling the dashboard and publishing the Story. The schedule below is a planning structure; actual sprint completion dates and velocity were not independently supplied.

Sprint	Focus	Planned tasks / deliverables	Estimate
Sprint 1	Data understanding	Connect CSV, review fields and types, prepare KPI cards and country maps.	20 story points
Sprint 2	Trend analysis	Create yearly petrol, diesel, LPG and Brent crude trend views; configure available filters.	20 story points
Sprint 3	Tax and comparison	Create regional tax chart, Top 5 tax-country ranking and three-country comparison; assemble dashboard.	20 story points
Sprint 4	Story and validation	Arrange seven Story points, review labels/aggregations and publish on Tableau Public.	20 story points

Planning assumption: 20 story points per six-day sprint gives a planned average of approximately 3.33 story points per day. This is an estimate from the project plan, not a measured historical velocity.

Key milestones

- Data connected and required fields identified.
- Core KPI cards and country maps prepared.
- Yearly trends and taxation comparison views completed.
- Dashboard assembled and checked.
- Seven-point Story published on Tableau Public.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Testing focuses on whether the Tableau views present the intended measures and comparisons. Performance metrics that were not captured are explicitly marked as not recorded rather than estimated.

Test area	Project result / evidence status
Data rendered	CSV source: global_fuel_prices_Dataset.csv. Previously reported project scope: 27,468 records, 84 countries, 7 regions and years 2020–2026. CSV file size and measured load time were not recorded.
Data preprocessing	Relevant Country, Region, Year, fuel-price, Brent crude and tax fields are used. Null/duplicate counts and detailed cleaning outcomes need source-data evidence.
Filters	Country and Year filters are used where configured. No timed filter-response comparison was recorded.
Calculations	AVG of petrol, diesel, LPG, Brent crude and tax percentage; yearly trend views; regional grouping; descending sort and Top 5 filter.
Dashboard	Four KPI cards, two country maps, four yearly trend charts, one regional tax pie chart, one Top 5 tax-country bar chart and one three-country comparison view. The count is 12 if each KPI card is counted as a visual element.
Story	Seven story points, progressing from maps to trends, tax analysis, country comparison and conclusion.
Query history	No Snowflake SQL query history exists for this CSV-to-Tableau implementation. Tableau query execution times were not recorded.

Evidence to attach

- Screenshot of the Tableau Data Source page showing the connected CSV and field list.
- Screenshot of KPI cards and both country maps.
- Screenshot of the yearly trend charts and regional tax chart.
- Screenshot of the Top 5 tax-country chart and three-country comparison.
- Screenshot of the published dashboard and the seven-point Tableau Story.

7. RESULTS

7.1 Output Screenshots

The implemented Tableau project contains the following output views. Insert screenshots captured directly from your Tableau Public workbook in the spaces below before final submission.

Output 1 – KPI Summary

Insert screenshot showing average diesel price, average LPG price, average petrol price and average tax percentage.

Output 2 – Petrol Price Map

Insert screenshot of the country-wise average petrol price map.

Output 3 – Diesel Price Map

Insert screenshot of the country-wise average diesel price map.

Output 4 – Yearly Trends

Insert screenshot of petrol, diesel, LPG and Brent crude yearly trend charts.

Output 5 – Taxation Analysis

Insert screenshot of average tax percentage by region and the Top 5 tax-country bar chart.

Output 6 – Country Comparison

Insert screenshot comparing petrol, diesel and LPG prices in Canada, Mexico and the United States.

Output 7 – Tableau Story

Insert screenshot showing the seven-point Story navigation and conclusion.

Published Tableau Public project: **Open the live Story**

8. ADVANTAGES & DISADVANTAGES

Advantages

- Combines multiple fuel and taxation comparisons in one report.
- Maps make country-level differences easier to identify.
- Yearly line charts help users follow changes over time.
- KPI cards provide a quick summary of the selected measures.
- Tableau Public makes the published Story easy to share through a browser link.
- The Story format guides viewers through the analysis in a logical order.

Disadvantages / limitations

- The analysis depends on the completeness, accuracy and coverage of the source CSV.
- Average values can hide variation within a country or across shorter time periods.
- The project is descriptive; it does not provide a validated forecasting model.
- Measured load times, filter response times and detailed data-quality counts were not captured.
- The current implementation uses Tableau Public and a CSV source; it does not include a live database pipeline or automated refresh process.

9. CONCLUSION

The Global Fuel Prices and Taxation Trends project demonstrates how a multi-country CSV dataset can be transformed into an accessible visual analytics report using Tableau Public. By combining KPI cards, country maps, yearly trend charts, regional tax analysis, a Top 5 ranking and a focused three-country comparison, the project gives users several ways to explore fuel-price and taxation patterns.

The seven-point Tableau Story provides a guided route through the analysis, moving from geographic differences to time trends and taxation comparisons before presenting a conclusion. The project meets its central objective of making the dataset easier to explore and compare, while remaining clear about the limits of the available data and performance evidence.

Key takeaway: one dataset, multiple perspectives — geography, time, fuel type and taxation — brought together in one Tableau Public experience.

10. FUTURE SCOPE

- Add more recent fuel-price observations and document the data refresh process.
- Include additional fuel types, countries or economic indicators where reliable data is available.
- Add verified data-quality checks with recorded null and duplicate counts.
- Capture and compare dashboard load and filter response times under consistent test conditions.
- Add validated forecasting or scenario analysis as a separate future enhancement.
- Improve mobile and small-screen layout after testing on target devices.
- Provide a downloadable data dictionary describing field names, units and calculation rules.

11. APPENDIX

A. Source Code (if any)

No separate source-code repository or custom application code was supplied for this project. The implemented analysis is contained in the Tableau Public workbook. Tableau calculated measures and worksheet configurations form the main analytical implementation.

B. Dataset Link

Dataset file referenced: **global_fuel_prices_Dataset.csv**. A public dataset download URL was not provided. Attach the CSV or provide its approved source URL when submitting the report.

C. GitHub & Project Demo Link

GitHub: Not provided / not applicable to the current Tableau-only implementation.

Project demo / published Tableau Story:
https://public.tableau.com/app/profile/ansh.sharma1711/viz/Myfirstproject_17931306181990/Story1

D. Notes on project evidence

Dataset scope figures (27,468 records, 84 countries, 7 regions and 2020–2026) are carried forward from the earlier project brief. Confirm them against the actual CSV before final submission. Numerical chart findings, null/duplicate counts, file size and measured performance times have not been invented. Add original Tableau screenshots in Section 7.

— END OF REPORT —